

# Clinical Protocol for Community-Acquired Pneumonia (CAP)

## 1. Diagnosis and Severity Assessment (CURB-65)

Community-Acquired Pneumonia is characterized by symptoms such as cough, fever, dyspnea, and pleuritic chest pain, confirmed by a new infiltrate on a chest radiograph.

Triage patients using the CURB-65 Scoring System (1 point for each):

- C: Confusion (new disorientation in person, place, or time).
- U: Urea > 7 mmol/L (Blood Urea Nitrogen > 19 mg/dL).
- R: Respiratory Rate  $\geq$  30 breaths per minute.
- B: Blood Pressure (Systolic < 90 mmHg or Diastolic  $\leq$  60 mmHg).
- 65: Age  $\geq$  65 years.

Triage Guidelines:

- Score 0-1: Low risk, treat as outpatient.
- Score 2: Moderate risk, consider short-stay inpatient admission or close monitoring.
- Score 3-5: High risk, urgent inpatient hospital admission; if score is 4-5, evaluate for Intensive Care Unit (ICU) admission.

## 2. Empirical Antibiotic Regimens

Outpatient Treatment (No comorbidities):

- Amoxicillin 1g three times daily OR Doxycycline 100mg twice daily OR Macrolide (Azithromycin 500mg day 1, then 250mg daily) in areas with macrolide resistance < 25%.

Outpatient Treatment (With comorbidities like chronic heart/lung/liver/renal disease, diabetes, alcoholism):

- Combination therapy: Amoxicillin/Clavulanate (875/125mg twice daily) PLUS a Macrolide (Azithromycin 500mg daily) OR Doxycycline 100mg twice daily.
- Alternative: Respiratory Fluoroquinolone (Levofloxacin 750mg daily or Moxifloxacin 400mg daily).

Inpatient Treatment (Non-Severe):

- Beta-lactam (Ceftriaxone 1-2g daily or Ampicillin-Sulbactam 1.5-3g every 6 hours) PLUS a Macrolide (Azithromycin 500mg daily) OR Respiratory Fluoroquinolone monotherapy.

## 3. Supportive Care and Follow-up

Supportive Care:

- Supplemental oxygen to maintain oxygen saturation  $\geq$  92% (or 88-92% in patients with COPD).
- Adequate hydration (intravenous fluids if patient is unable to tolerate oral intake).
- Antipyretics (Acetaminophen or Ibuprofen) for fever control.

Follow-up:

- Clinical assessment within 48-72 hours of starting antibiotics to monitor treatment response.
- Repeat chest radiograph in 4-6 weeks for patients over 50 years or smokers to confirm resolution and rule out underlying malignancy.